# Hypothesis Testing with E-Values

## Chapter 1: Introduction

Based on Ramdas & Wang

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# Chapter Overview

# Notation You'll Need for This Chapter I

Before diving in, here is a plain-English glossary of every non-elementary term and symbol used later in this deck. Nothing here requires measure theory — treat each analogy as the working definition you need.

- **Null hypothesis,** $\mathcal{P}$: the "boring" explanation of the data we test against – a *set* of candidate probability distributions (e.g., "the coin is fair," i.e., $\{\text{Bernoulli}(0.5)\}$).
- **Alternative hypothesis,** $\mathcal{Q}$: the "interesting" explanation we'd like evidence for (e.g., "the coin is biased towards heads").
- $\mathbb{P}, \mathbb{Q}$: a single probability distribution; $\mathbb{P} \in \mathcal{P}$ means $\mathbb{P}$ is one specific member of the null hypothesis set.
- **Simple vs. composite hypothesis**: simple means the set has exactly one distribution (e.g., $\{\mathbb{P}\}$); composite means it has more than one (e.g., "$\theta \neq 0.5$" covers infinitely many biased coins).
- **Random variable** $X$: a quantity whose value depends on the random outcome (here: our data).
- $\mathbb{E}^{\mathbb{P}}[X]$: the expectation (long-run average) of $X$, computed *assuming* $\mathbb{P}$ is the true distribution.

# Notation You'll Need for This Chapter II

- **E-variable / e-value**: a nonnegative random variable $E$ with $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ under every null distribution. Analogy: a bet that is never favorable to you (on average) if the null is true, but pays off big if the null is false. "E-value" is the number you actually observe.
- **P-variable / p-value**: you already know this one — a number in $[0, 1]$ that is uniform-or-larger under the null. Small p-value $=$ evidence against the null. (*Opposite* convention to e-values: for e-values, *large* is evidence against the null.)
- **Test** $\phi$: a (possibly randomized) decision rule taking values in $[0, 1]$, interpreted as "the probability I reject $H_0$." A **binary** test only ever outputs 0 ("don't reject") or 1 ("reject").
- **Type-I error / level** $\alpha$: the chance of a false alarm (rejecting a true null). A "level-$\alpha$ test" controls this chance to be at most $\alpha$ (e.g., $\alpha = 0.05$).
- **Likelihood ratio** $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$: for a data point $x$, how much more likely $x$ is under $\mathbb{Q}$ than under $\mathbb{P}$. If data are continuous/discrete with densities (or probability mass functions) $q(x), p(x)$, this is just $q(x)/p(x)$.

# Notation You'll Need for This Chapter III

- **Exact e/p-variable**: one where the defining inequality is always an equality ($\mathbb{E}^{\mathbb{P}}[E] = 1$, or $\mathbb{P}(P \leq \alpha) = \alpha$) — the "tightest possible" case.
- **Stochastically larger**: random variable $X$ is stochastically larger than a reference distribution if $X$ is, loosely, "at least as likely to be big." (Formally: its cdf is never above the reference cdf.)
- **Supermartingale idea (informal)**: a wealth process that, on average, never grows from one step to the next given the past — like a casino game that is fair or tilted against you. We will only need the intuition, not the formal machinery.
- **Filtration / adapted / stopping time (informal)**: think of a filtration as "the information you've seen up to now, growing over time"; a process is adapted if, at each time, it only uses information seen so far; a stopping time is a rule for "when to stop" that only looks at data observed up to that moment (no peeking into the future).

# Notation You'll Need for This Chapter IV

- **E-process**: a sequence of e-values $E_1, E_2, \ldots$ (built up as data arrive) that remains a valid e-value *even if you stop early*, at any data-dependent stopping time. This is what lets e-values handle "peeking" safely.
- **Exchangeable**: a collection of random variables whose joint behavior does not change if you relabel/permute them – e.g., shuffling a deck of otherwise-identical-looking cards.
- **Calibrator**: a fixed recipe that converts one kind of statistic into another while keeping validity, e.g., turning an e-value $E$ into a p-value via $P = 1/E$.
- **Log-optimal e-value / e-power**: among all valid e-values, the one that maximizes $\mathbb{E}^{\mathbb{Q}}[\log E]$ (expected log growth rate under the alternative $\mathbb{Q}$) – explained with concrete intuition in Section 1.2.
- **Markov's inequality (used implicitly)**: for any nonnegative random variable $E$ with $\mathbb{E}[E] \leq 1$, we automatically get $\mathbb{P}(E \geq 1/\alpha) \leq \alpha$ – this is exactly why "reject when $E > 1/\alpha$" is a valid level-$\alpha$ test.

# 1.1 The Three Roles of E-Values I

An e-value (nonnegative statistic with expectation at most 1 under the null; reminder from the notation frame) plays three distinct roles in the book. Understanding these upfront will help you see why so many later chapters revisit the same simple object.

### Defining property (restated)

An e-value is a nonnegative test statistic whose expected value is at most one under the null hypothesis.

- "e" can stand for *expectation* (its defining constraint) or *evidence* (how we use it).
- Larger e-value $\Rightarrow$ more evidence *against* the null – the opposite direction from p-values.

# 1.1 The Three Roles of E-Values II

1. **Methodological.** E-values directly yield familiar tools: the reciprocal $1/E$ of an e-value is a valid p-value, and rejecting the null when $E$ exceeds $1/\alpha$ is a valid level-$\alpha$ test (both via Markov's inequality). E-values also unlock things p-values struggle with: universal inference for irregular problems, log-optimal ("uniformly most e-powerful") e-values, sequential anytime-valid testing, and simple false-discovery-rate control.
2. **Technical.** E-values show up as intermediate tools even in problems that don't look like they involve e-values at all – e.g., every procedure that controls the false discovery rate turns out to be an instance of an e-value-based procedure in disguise. They are also intimately tied to nonnegative (super)martingales (the "fortune process never grows on average" idea from the notation frame).
3. **Fundamental.** At the deepest level: a powered test exists for a testing problem if and only if a powered e-variable exists. E-values are not just a convenience – they are equivalent, foundational objects to hypothesis tests themselves.

The book positions itself as building *a theory of evidence* (via e-values), complementing the classical *theory of decision-making* (Neyman, Pearson, Wald, Savage).

# 1.2.1 P-values and E-values Are Statistical Cousins I

The book does *not* argue e-values should replace p-values. Instead, it argues both deserve equal footing as complementary tools. Why call them “cousins”? Because they can be converted into each other while preserving key guarantees.

# 1.2.1 P-values and E-values Are Statistical Cousins II

## E-to-p and p-to-e calibration

The intuition: an e-value's reciprocal behaves like a p-value, and an indicator of "was the p-value small" behaves like an e-value.

$$\text{(e-to-p)} \quad P = \frac{1}{E} \qquad\qquad \text{(p-to-e, at level } \alpha\text{)} \quad E = \frac{\mathbf{1}\{P \le \alpha\}}{\alpha}.$$

- $E$: an e-variable for $\mathcal{P}$; $P$: a p-variable for $\mathcal{P}$.
- $\mathbf{1}\{P \le \alpha\}$: the indicator function, $1$ if $P \le \alpha$ is true, else $0$.
- e-to-p calibration *preserves the rejection decision exactly*: for any $\alpha$, $1/E \le \alpha \iff E \ge 1/\alpha$.
- The specific p-to-e calibrator above also preserves the decision at that fixed level $\alpha$ (it is the "all-or-nothing" e-value).

# 1.2.1 P-values and E-values Are Statistical Cousins III

**Key differences** (why you might prefer one over the other):

- **Post-hoc decisions.** With e-values, the rejection level $\alpha$ can be chosen *after* seeing the data, and the guarantee still holds. This is provably false for p-values in general.
- **Optional continuation.** If several scientists sequentially run follow-up experiments, their e-values can always be *merged by simple multiplication* – even if later experiments were only run because earlier ones looked promising. Sequentially-obtained p-values can be very hard to combine correctly in this setting.
- **Unknown sampling scheme.** P-values need a fully specified data-collection design to be valid; a specific family of e-values remains valid regardless of the (reasonable) stopping/design rule used – illustrated concretely in Example 1.4 (Section 1.6).
- **Multiple testing.** "Compound" e-values turn out to be central to multiple-testing procedures (Chapters 9, 11, 13) in ways that have no clean p-value analogue.

# 1.2.2 Power and E-power I

A natural question: *are e-values more or less powerful than p-values?* The book's answer is subtle – power is a property of a *procedure* (a threshold test), not of e-values or p-values in isolation.

- If a p-value $P$ is derived from an e-value $E$ via $P = 1/E$, the two thresholded tests are identical – no difference in power.
- If an e-value is derived from a p-value via $E = \mathbf{1}\{P \leq \alpha\}/\alpha$, the resulting level-$\alpha$ test again has exactly the same power as the original p-value test.
- Otherwise (general e-values vs. general p-values), no universal comparison is possible – it depends on which specific e-value or p-value you picked.

# 1.2.2 Power and E-power II

## E-power: the criterion the book actually optimizes

Intuition first: if we care about the null hypothesis being *wrong* (alternative $\mathbb{Q}$ true), we'd like our e-value to grow as fast as possible under $\mathbb{Q}$. Taking the expected *logarithm* (rather than the expectation) turns out to be the right notion of "fast growth" when e-values will be multiplied together across repeated experiments.

$$\text{e-power of } E \text{ against } \mathbb{Q} \;:=\; \mathbb{E}^{\mathbb{Q}}[\log(E)].$$

- $\mathbb{E}^{\mathbb{Q}}[\cdot]$: expectation computed as if the alternative $\mathbb{Q}$ generated the data.
- $\log$: natural logarithm.
- An e-value that maximizes e-power is called **log-optimal**.

# 1.2.2 Power and E-power III

- Why the log? If scientists sequentially multiply e-values $E_1, E_2, \ldots$ (valid even under data-dependent stopping/continuation), then $\prod_{i=1}^{n} E_i = \exp(n \cdot \overline{\log E}_{[n]})$, so the *rate* at which the product diverges to $\infty$ is governed exactly by $\mathbb{E}[\log E_i]$.
- Log-optimality is therefore not an arbitrary choice of utility function – it is forced by the structure of repeated multiplication.

# 1.2.2 Power and E-power IV

One subtlety worth remembering: e-values in this book are chosen to be log-optimal, *not* to maximize power at a fixed level $\alpha$.

- A power-maximizing e-value degenerates into an "all-or-nothing" e-value $\mathbf{1}_A/\alpha$ for some rejection set $A$ – but this has e-power $= -\infty$ (it equals zero with positive probability, and $\log(0) = -\infty$).
- Log-optimal e-values, by contrast, are always strictly positive (with probability 1 under $\mathbb{Q}$), so they never risk this catastrophic penalty.

# 1.2.2 Power and E-power V

- **Takeaway:** if you plug a log-optimal e-value directly into a decision procedure built around classical power, you may be disappointed – not because e-values are weak, but because you're comparing objects optimized for different criteria.

# Python: E-to-P and P-to-E Calibration, Checked Numerically

```python
import numpy as np
rng = np.random.default_rng(0)

alpha = 0.05
n_reps = 200_000

# Start from a valid e-value E (mean <= 1 under H0) for our coin example:
# H0: theta=0.5 (fair coin), n=20 flips, LR e-value vs theta=0.7
n, p0, q0 = 20, 0.5, 0.7
S = rng.binomial(n, p0, size=n_reps)                # data simulated under H0
E = (q0/p0)**S * ((1-q0)/(1-p0))**(n-S)              # e-value (eq. 1.5)

P_from_E = 1.0 / E                                   # e-to-p calibration
print("mean(E) under H0          =", E.mean())        # should be <= 1
print("P(P_from_E <= alpha)      =", (P_from_E <= alpha).mean(),
      " vs alpha =", alpha)                           # should be <= alpha
print("P(E >= 1/alpha)           =", (E >= 1/alpha).mean())  # identical event

# Now go the other way: turn a p-value into an e-value at level alpha
E_from_P = (P_from_E <= alpha).astype(float) / alpha  # p-to-e calibration
print("mean(E_from_P) under H0 =", E_from_P.mean())    # should be <= 1
```

**Output (excerpt):** mean(E) $\approx$ 0.982, P(P_from_E $\leq$ 0.05) $\approx$ 0.006 $\leq$ 0.05, mean(E_from_P) $\approx$ 0.006 $\leq$ 1 — exactly the two calibration guarantees from the block above, confirmed by simulation.

# 1.3 Mathematical Notation and Conventions I

This section fixes the formal scaffolding used throughout the book. We only need the parts relevant to the definitions in the next section – treat the rest as background color.

- Data live in a sample space $\Omega$ (all possible outcomes), equipped with a $\sigma$-algebra $\mathcal{F}$ (informally: "all the events we're allowed to assign a probability to"; see notation frame).
- $\mathcal{M}_1 =$ the set of *all* probability distributions on $(\Omega, \mathcal{F})$. Its elements are also just called "distributions."
- A random variable $X$ is a function from $\Omega$ to $\mathbb{R}$ (or $[-\infty, \infty]$ if we explicitly allow infinite values).
- The true (but unknown) distribution generating our data $X$ is $\mathbb{P}^\dagger \in \mathcal{M}_1$. Data can be a vector $X = (X_1, \ldots, X_n)$, possibly (but not necessarily) i.i.d.
- $X^t := (X_1, \ldots, X_t)$: shorthand for the first $t$ data points.

# 1.3 Mathematical Notation and Conventions II

## Definition 1.1: Hypothesis

*Intuition:* a hypothesis is simply the collection of distributions that are "consistent with" a scientific claim. If the claim pins down exactly one distribution, there's no remaining uncertainty about which $\mathbb{P}$ is true (simple); otherwise several distributions are all compatible with the claim (composite).

A *hypothesis* is a set of probability measures in $\mathcal{M}_1$. A hypothesis is *simple* if it is a singleton, such as $\{\mathbb{P}\}$ and $\{\mathbb{Q}\}$. Otherwise it is *composite*.

- We always write $\mathcal{P}$ for the null hypothesis and $\mathcal{Q}$ for the alternative hypothesis; $H_0$ / $H_1$ denote the corresponding statements "$\mathbb{P}^\dagger \in \mathcal{P}$" / "$\mathbb{P}^\dagger \in \mathcal{Q}$."
- $\mathcal{P}, \mathcal{Q}$ are always assumed non-intersecting (they can't both be true).
- $\mathrm{Conv}(\mathcal{P})$: the convex hull of $\mathcal{P}$ (all probability-weighted mixtures of distributions in $\mathcal{P}$).

# 1.3 Mathematical Notation and Conventions III

- $\mathbb{Q}$ is *absolutely continuous* with respect to $\mathbb{P}$, written $\mathbb{Q} \ll \mathbb{P}$, if $\mathbb{P}(A) = 0 \Rightarrow \mathbb{Q}(A) = 0$ for every measurable set $A$. Intuition: anything $\mathbb{P}$ deems impossible is also impossible under $\mathbb{Q}$, so it makes sense to compare their densities pointwise.
- When $\mathbb{Q} \ll \mathbb{P}$, we write $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ for the likelihood ratio (reminder from the notation frame): if $q, p$ are densities/pmfs of $\mathbb{Q}, \mathbb{P}$, then $(\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P})(x) = q(x)/p(x)$.
- $X$ is *stochastically larger* than a distribution with cdf $F$ if $\mathbb{P}(X \leq x) \leq F(x)$ for all $x \in \mathbb{R}$ (reminder: "at least as likely to be big").
- $\mathbb{N} = \{1, 2, \dots\}$; $\mathbb{R} = (-\infty, \infty)$; $[K] := \{1, \dots, K\}$; $\Delta_K$ is the probability simplex in $\mathbb{R}^K$: $\Delta_K = \{(v_1, \dots, v_K) \in [0,1]^K : \sum_{k=1}^K v_k = 1\}$.
- The Euler number $\exp(1)$ is denoted $\mathrm{e}$ (upright) to avoid clashing with the variable $E$ used for e-values.
- "Increasing" / "decreasing" are always meant in the *non-strict* sense; $x \wedge y = \min(x, y)$, $x \vee y = \max(x, y)$.

# 1.4 Definitions of E-variables, P-variables, and Tests I

Here is the formal heart of the chapter. Before the wall of symbols: an e-variable is a "fair-or-worse bet against the null," a p-variable is the classical p-value object, and a test is a (possibly randomized) accept/reject rule. Let $\mathcal{P} \subseteq \mathcal{M}_1$ be the null hypothesis.

**Definition 1.2: E-variables, P-variables, and Tests**

(i) An *e-variable* $E$ for $\mathcal{P}$ is a $[0, \infty]$-valued random variable satisfying $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ for all $\mathbb{P} \in \mathcal{P}$. It is *exact* if $\mathbb{E}^{\mathbb{P}}[E] = 1$ for all $\mathbb{P} \in \mathcal{P}$.

(ii) A *p-variable* $P$ for $\mathcal{P}$ is a $[0, \infty)$-valued random variable satisfying $\mathbb{P}(P \leq \alpha) \leq \alpha$ for all $\alpha \in (0, 1)$ and all $\mathbb{P} \in \mathcal{P}$. It is *exact* if $\mathbb{P}(P \leq \alpha) = \alpha$ for all $\alpha, \mathbb{P}$.

(iii) A *test* $\phi$ is a $[0, 1]$-valued random variable. It is *binary* if its range is $\{0, 1\}$. The *type-I error (rate)* of $\phi$ for $\mathbb{P}$ is $\mathbb{E}^{\mathbb{P}}[\phi]$. A test has *level* $\alpha \in [0, 1]$ for $\mathcal{P}$ if its type-I error is at most $\alpha$ for every $\mathbb{P} \in \mathcal{P}$.

- $\mathfrak{E} = \mathfrak{E}(\mathcal{P})$: the set of *all* e-variables for $\mathcal{P}$; $\mathfrak{U} = \mathfrak{U}(\mathcal{P})$: the set of all p-variables for $\mathcal{P}$.

# 1.4 Definitions of E-variables, P-variables, and Tests II

**Remarks worth internalizing:**

1. A test allows external randomization: draw $U \sim \mathrm{Unif}(0,1)$ independent of the data, reject if $U \leq \phi$. Then $\mathbb{E}^{\mathbb{P}}[\phi]$ is exactly the unconditional rejection probability.
2. Equivalently, $P$ is a p-variable iff $P$ is stochastically larger than $\mathrm{Unif}(0,1)$ under every $\mathbb{P} \in \mathcal{P}$.
3. A large realized e-value is evidence *against* $H_0$ (because its mean is capped at 1 under the null, so a big value is "surprising"); a small realized p-value is likewise evidence against $H_0$.
4. We say "an e-variable for $H_0$" to mean "for $\mathcal{P}$."
5. Most tests we use are binary, with $\phi_0 \equiv 0$ (never reject) as the trivial level-0 test.
6. An e-variable is allowed to equal $\infty$ (licence to reject outright), but only on *null-sets* (events impossible under every $\mathbb{P} \in \mathcal{P}$) – otherwise $\mathbb{E}^{\mathbb{P}}[E]$ couldn't stay $\leq 1$. The constant $E \equiv 1$ is always a valid (if useless) e-variable for any $\mathcal{P}$.

# 1.4 Definitions of E-variables, P-variables, and Tests III

## Fact 1.3: Useful closure properties

*Intuition:* these are the "legal moves" you're allowed to make with e- and p-variables without breaking validity – e.g., you can always average several e-values together, or multiply *independent* ones, and the result is still a legitimate e-variable. Let $F_{X|\mathbb{P}}$ be the cdf of $X$ under $\mathbb{P}$.

1. For any $X, \mathbb{P}$: $F_{X|\mathbb{P}}(X)$ is a p-variable for $\mathbb{P}$.
2. $\sup_{\mathbb{P} \in \mathcal{P}} F_{X|\mathbb{P}}(X)$ is a p-variable for $\mathcal{P}$.
3. If $E_1, \ldots, E_K$ are e-variables for $\mathcal{P}$, their **arithmetic average** is an e-variable for $\mathcal{P}$.
4. If $E_1, \ldots, E_K$ are e-variables for $\mathcal{P}$ and **independent** under each $\mathbb{P} \in \mathcal{P}$, their **product** is an e-variable for $\mathcal{P}$.
5. If $P$ is a p-variable and $P' \geq P$, then $P'$ is a p-variable.
6. If $E$ is an e-variable and $0 \leq E' \leq E$, then $E'$ is an e-variable.

# 1.4 Definitions of E-variables, P-variables, and Tests IV

## E-process (informal definition)

*Intuition:* an e-process is a whole time series of e-values that stays trustworthy *even if you decide when to stop looking based on the data itself* – crucial for sequential science.
A finite or infinite sequence $E_1, E_2, \ldots$ of e-variables (adapted to an underlying filtration; reminder: "only uses information seen so far") is called an *e-process* if, for every stopping time $\tau$ (a rule for stopping that doesn't peek into the future), $E_\tau$ is again a valid e-value.

- This innocuous-looking definition hides a deep connection to nonnegative (super)martingales, explored fully in Chapter 7.
- It is exactly the property that makes "optional continuation" (Section 1.2.1) and "unknown sampling schemes" (Example 1.4) work.

# Python: Checking Fact 1.3 (Averaging and Multiplying E-values)

```python
import numpy as np
rng = np.random.default_rng(1)
n, p0, n_reps = 20, 0.5, 100_000
S = rng.binomial(n, p0, size=n_reps)       # data simulated under H0

# Two e-variables for the SAME null, vs two different alternatives
E1 = (0.7/p0)**S * (0.3/(1-p0))**(n-S)    # LR e-value vs theta=0.7
E2 = (0.6/p0)**S * (0.4/(1-p0))**(n-S)    # LR e-value vs theta=0.6
E_avg = (E1 + E2) / 2                      # Fact 1.3(iii): average
print("mean(E_avg) =", E_avg.mean())       # <= 1 ?

# Product of e-variables from INDEPENDENT rounds (Fact 1.3(iv)):
Sa = rng.binomial(10, p0, size=n_reps)    # first 10 flips
Sb = rng.binomial(10, p0, size=n_reps)    # independent next 10 flips
Ea = (0.7/p0)**Sa * (0.3/(1-p0))**(10-Sa)
Eb = (0.7/p0)**Sb * (0.3/(1-p0))**(10-Sb)
print("mean(Ea*Eb) =", (Ea * Eb).mean())  # <= 1 ?
```

**Output:** both means come out close to $1$ (typically $0.97$–$1.00$) and never exceed it, confirming Fact 1.3(iii)–(iv).

## 1.5 Betting Interpretation of E-values

E-variables have a clean interpretation as *fair or unfavorable bets* against the null. This framing is the intuitive engine behind most of the book's constructions.

### The Forecaster/Skeptic game

*Intuition:* Skeptic is trying to make money betting against Forecaster's claim that $\mathbb{P} \in \mathcal{P}$ describes the world. An e-variable is precisely a bet that Forecaster (a rational, expected-value-maximizing party) would be willing to accept.

- Skeptic pays \$1 up front and proposes a payout function $E : \Omega \to \mathbb{R}_+$: if outcome $\omega$ occurs, Forecaster pays Skeptic $E(\omega)$ dollars.
- Forecaster accepts the contract only when it is favorable to them – i.e., when $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ (Skeptic doesn't expect to profit if $\mathbb{P}$ is really true). This is *exactly* the e-variable condition.
- If Skeptic secretly believes $\mathbb{Q}$ is the truth, they will only offer this particular $E$ if $\mathbb{E}^{\mathbb{Q}}[E] > 1$ – i.e., they expect to come out ahead.

A large realized e-value $E(\omega)$ = Skeptic made a lot of money = strong evidence that Forecaster's null was wrong.

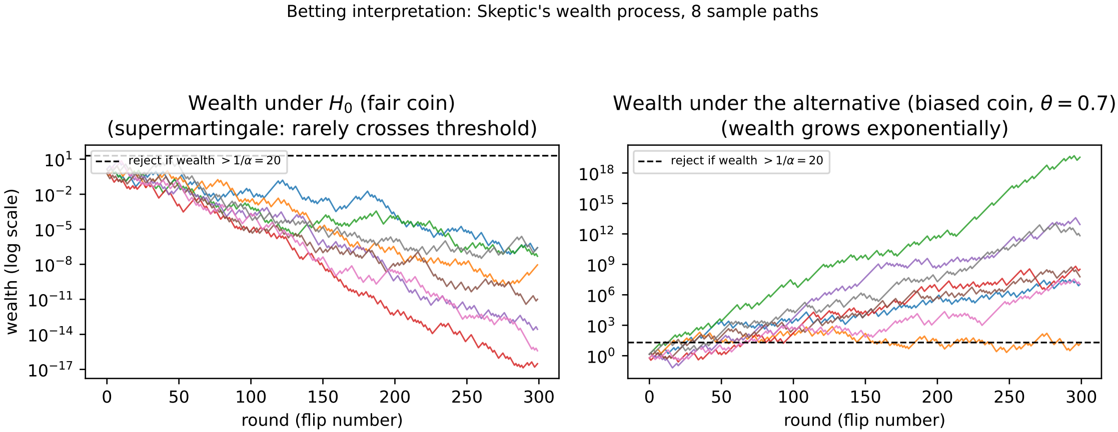
# Contrast: The (Flawed) Betting Interpretation of P-values

P-values have a betting interpretation too – but a flawed one, which helps explain why e-values are the more natural currency for combining evidence across experiments.

- For every $r \in (0, 1]$, Forecaster offers a contract priced at \$1 that pays $1/r$ dollars if statistic $P < r$, and nothing otherwise.
- Forecaster only offers this if $P$ is a p-variable for $\mathcal{P}$ (equivalently, $\mathbf{1}\{P \leq r\}/r$ is an e-variable for any fixed $r$).
- But here, the bets are indexed by $r$, and the *realized* p-value equals the smallest $r$ at which Skeptic *would have* profited, **had they picked that particular bet in advance**.
- This is "double dipping": p-values implicitly report the best bet in hindsight, whereas an e-value is a single, honestly pre-committed bet. That is why e-values and $1/(\text{p-value})$ should never be compared on the same numeric scale.

This distinction is exactly what powers the "optional continuation" and "post-hoc decision" advantages of Section 1.2.1.

# Figure: The Betting Wealth Process




Eight simulated wealth paths from repeatedly betting $E = (0.7/0.5)^{\text{heads}}(0.3/0.5)^{\text{tails}}$ per flip. **Left:** under the null ($\theta = 0.5$), wealth fluctuates but rarely exceeds the rejection threshold $1/\alpha$ (supermartingale intuition: on average, wealth does not grow). **Right:** under the true alternative ($\theta = 0.7$), wealth grows exponentially – Skeptic profits.

# Python: Simulating the Wealth Process and Ville's Inequality

```python
import numpy as np
rng = np.random.default_rng(2024)

alpha = 0.05
threshold = 1/alpha                 # reject if wealth ever exceeds this
n_rounds = 300
n_reps = 5000
p0, q0 = 0.5, 0.7                   # H0: fair coin; bet built against theta=0.7

max_wealth = np.zeros(n_reps)
for r in range(n_reps):
    flips = rng.binomial(1, p0, size=n_rounds)          # data under H0
    per_flip_e = np.where(flips == 1, q0/p0, (1-q0)/(1-p0))
    wealth = np.cumprod(per_flip_e)                      # running product = wealth
    max_wealth[r] = wealth.max()

frac_exceeds = np.mean(max_wealth > threshold)
print(f"P(max wealth over {n_rounds} rounds > 1/alpha={threshold:.0f}) = {frac_exceeds:.4f}")
print(f"Ville's inequality guarantees this is <= alpha = {alpha}")
```

**Output:** `P(max wealth > 20) = 0.0490`, comfortably respecting the $\leq 0.05$ guarantee – even though we checked the *running maximum* over all 300 rounds, not just the final value. This is the practical payoff of the e-process property.

# 1.6 A First Example: Likelihood Ratio

The most natural e-variable of all: for testing a simple null $\mathbb{P}$ against a simple alternative $\mathbb{Q} \ll \mathbb{P}$ (reminder: absolutely continuous, i.e., comparable densities), the likelihood ratio itself is an e-variable.

## Likelihood-ratio e-variable

*Intuition:* how much more plausible is the data under $\mathbb{Q}$ than under $\mathbb{P}$? That ratio, applied to the observed data, is automatically a valid (exact) e-value for $\mathbb{P}$.

$$E = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(X).$$

- $X$: the observed data.
- For *any* $\mathbb{Q} \ll \mathbb{P}$, this $E$ is an e-variable for $\mathbb{P}$, and in fact *exact* ($\mathbb{E}^{\mathbb{P}}[E] = 1$).

We specialize to two settings the book uses throughout: Gaussian mean shift, and our running Bernoulli (coin flip) example.

## Gaussian example (Eq. 1.1–1.2)

Testing $\mathbb{P} = \mathrm{N}(0,1)$ against $\mathbb{Q} = \mathrm{N}(\mu,1)$ ($\mu > 0$ known), with i.i.d. data $X_1, \ldots, X_n$ and $S_n = \sum_i X_i$:

$$E_n = \exp\big(\mu S_n - n\mu^2/2\big) = \prod_{i=1}^{n} \exp(\mu X_i - \mu^2/2).$$

- $S_n$: sum of the $n$ observations.
- Each factor $\exp(\mu X_i - \mu^2/2)$ is itself an exact e-variable; their product remains exact ($\mathbb{E}^{\mathbb{P}}[E_n] = 1$).

Writing $\delta = \mu\sqrt{n}$ (signal strength) and $Z = S_n/\sqrt{n}$ (standard normal under $H_0$):

$$E_n = \exp\big(\delta Z - \delta^2/2\big), \qquad P_n = \Phi(-S_n/\sqrt{n}) = \Phi(-Z)$$

- $\Phi$: standard normal cdf; $P_n$ is the classical Neyman–Pearson p-variable.
- $E_n$ increasing in $S_n$, $P_n$ decreasing in $S_n$: both agree that larger $S_n$ favors $\mathbb{Q}$.

# 1.6 Likelihood Ratio: the Two-Sided Version

## Two-sided version (Eq. 1.3–1.4)

Testing $\mathrm{N}(0,1)$ against $\mathrm{N}(\mu,1)$ for *unknown* $\mu \neq 0$, using a free parameter $\delta > 0$:

$$E_n = \frac{\exp(\delta Z - \delta^2/2) + \exp(-\delta Z - \delta^2/2)}{2}, \qquad P_n = 2\Phi(-|Z|).$$

- Averaging the "$+\mu$" and "$-\mu$" bets (Fact 1.3(iii)) gives a valid e-variable for the two-sided problem.
- $\delta$ is a design choice affecting power – addressed in later chapters.

# 1.6 Likelihood Ratio: the Bernoulli (Coin-Flip) Version

## Bernoulli / coin-flip version (Eq. 1.5) – our running example

Testing $\mathbb{P} = \mathrm{Bernoulli}(p)$ vs. $\mathbb{Q} = \mathrm{Bernoulli}(q)$, $q > p$, sample size $n$, $S_n = \sum_i X_i$ (number of heads):

$$E_n = \left(\frac{q}{p}\right)^{S_n} \left(\frac{1-q}{1-p}\right)^{n-S_n} .$$

- $E_n$ exact, increasing in $S_n$; Neyman–Pearson p-variable is $P_n = 1 - F_{n,p}(S_n)$ (upper binomial tail).
- This is our **running example** for the rest of the deck: testing whether a coin is fair.

# Worked Example: Is Our Coin Fair? (Step by Step)

**Setup.** $H_0 : \theta = 0.5$ (fair coin) vs. chosen alternative $\theta = 0.7$. We flip $n = 20$ times and observe $S_{20} = 14$ heads.

## Computing the e-value, one step at a time

- **Step 1** (identify the formula): use Eq. 1.5 with $p = 0.5$, $q = 0.7$, $n = 20$, $S_n = 14$.
- **Step 2** (plug in): $E_{20} = (0.7/0.5)^{14}\,(0.3/0.5)^6 = (1.4)^{14}(0.6)^6$.
- **Step 3** (compute each factor): $1.4^{14} \approx 111.12, \quad 0.6^6 = 0.046656$.
- **Step 4** (multiply): $E_{20} \approx 111.12 \times 0.046656 \approx 5.18$.
- **Step 5** (interpret): $E_{20} \approx 5.18 > 1$ is mild evidence *against* the fair-coin null – but well below $1/\alpha = 20$ for $\alpha = 0.05$, so we would *not* reject at that level.

**For comparison**, the exact one-sided p-value is $P(S_{20} \geq 14 \mid \theta = 0.5) \approx 0.058$ – also not significant at $\alpha = 0.05$, consistent with the e-value's verdict.

# Example 1.4: Unknown Sampling Scheme I

This is the book's own example illustrating why e-values tolerate unknown data-collection designs, while p-values do not.

## Setup

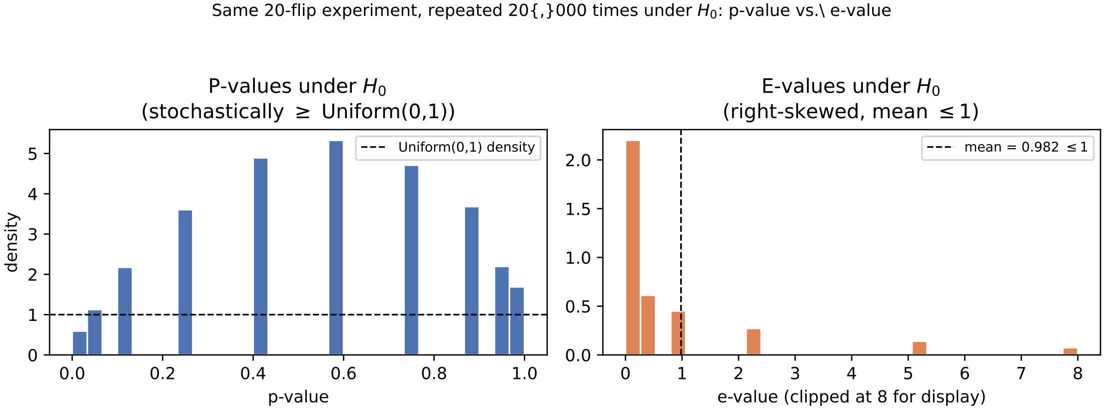
Testing $\mathbb{P} = \text{Bernoulli}(1/2)$ vs. $\mathbb{Q} = \text{Bernoulli}(q)$, $q > 1/2$. A scientist is handed the dataset $(1, 1, 0, 1, 1, 1, 1, 1)$ – 8 points, 7 heads – *without* being told how it was collected.

- **Design (a):** collect exactly 8 points. P-value $= \mathbb{P}(N_8 \geq 7) = 0.035$ (significant at 0.05).
- **Design (b):** collect data until 5 heads occur in a row. P-value $= \mathbb{P}(\text{5-in-a-row observed at or before } n = 8) = 0.078$ (*not* significant at 0.05).

# Example 1.4: Unknown Sampling Scheme II

- **The problem:** if only the data (not the design) is reported, the scientist cannot tell whether they're in case (a), (b), or some other design – yet the p-value's validity crucially depends on knowing which.
- **The fix:** the e-value $E_n = q^{S_n}(1-q)^{n-S_n}/2^n$ from Eq. 1.5 (with $p = 1/2$) remains a *valid* e-variable **regardless of the data-collecting design**, as long as all obtained data is used from the start.
- **Why:** $(E_n)_{n\geq 1}$ forms an e-process (Chapter 7) – it stays valid at any data-dependent stopping rule, which is exactly the "no peeking into the future" property from the notation frame.
- **Caveat:** this does not rescue *malicious* designs (cherry-picking only 1s, discarding early 0s, reporting only the "best" of several batches) – no statistical method purely based on collected data can fix that.

# Figure: P-values vs. E-values Under Repeated Null Experiments




20,000 simulated repeats of our $n = 20$ coin-flip experiment under $H_0 : \theta = 0.5$. **Left:** p-values are (stochastically) uniform on $[0, 1]$ – as required by the p-variable definition. **Right:** e-values are right-skewed (most realizations are modest or small, a few are large) with sample mean $\approx 0.98 \leq 1$, exactly as the e-variable definition demands. Numerically, $\mathbb{P}(E > 20) \approx 0.006 \leq 1/20$, confirming Markov's inequality.

# An Illustration: Sequential Data and Adaptive Strategies I

Suppose data $X_1, \dots, X_n$ from parameter $\theta^\dagger$ arrive one at a time, testing $H_0 : \theta^\dagger = 0$ vs. $H_1 : \theta^\dagger \in \Theta_1$. Let $\ell(x; \theta) = \mathrm{d}\mathbb{Q}_\theta / \mathrm{d}\mathbb{P}(x)$. Any choice of $\theta$ turns $\ell(X_i; \theta)$ into a valid e-variable for $H_0$ – so we have freedom in how to pick $\theta$ at each step:

(a) **Fixed**: pick $\theta_1, \dots, \theta_n \in \Theta_1$ in advance (possibly all equal).

(b) **Adaptive**: estimate $\theta_k$ from $X_1, \dots, X_{k-1}$ only – e.g., via maximum likelihood (MLE) or maximum a posteriori (MAP).

# An Illustration: Sequential Data and Adaptive Strategies II

## The running product is always a valid martingale

*Intuition:* no matter how cleverly (or adaptively) you pick each $\theta_i$, as long as you only use *past* data to do so, multiplying per-round e-variables together keeps you honest – your expected next-step wealth, given everything so far, is exactly your current wealth.
Define $M_n := \prod_{i=1}^{n} E_i$ with $E_i = \ell(X_i; \theta_i)$, $M_0 = 1$.

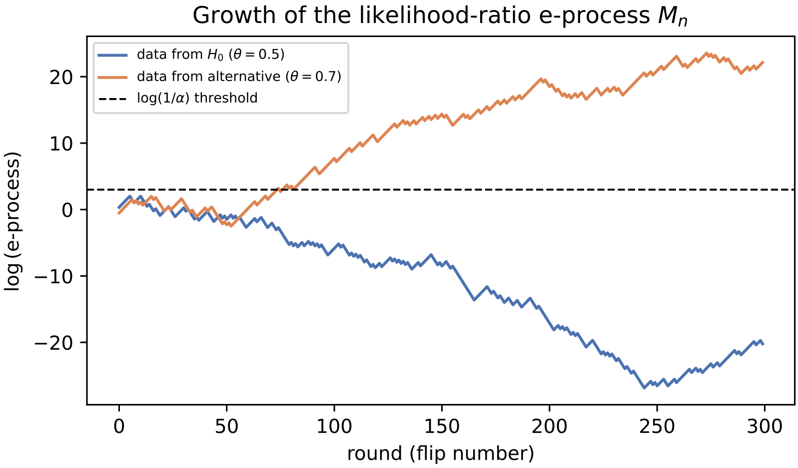
$$\mathbb{E}^{\mathbb{P}}\big[M_i \mid X_1, \ldots, X_{i-1}\big] = M_{i-1} \cdot \mathbb{E}^{\mathbb{P}}\big[\ell(X_i; \theta_i) \mid X_1, \ldots, X_{i-1}\big] = M_{i-1}.$$

- **Step 1**: factor $M_i = M_{i-1} \cdot E_i$ out of the conditional expectation ($M_{i-1}$ is already known given the past).
- **Step 2**: $\theta_i$ is computed from $X_1, \ldots, X_{i-1}$ only, so $\mathbb{E}^{\mathbb{P}}[\ell(X_i; \theta_i) \mid X_1, \ldots, X_{i-1}] = 1$ (it's an exact e-variable given the past).
- **Conclusion**: $(M_i)$ never grows *on average* given the past – the martingale/supermartingale property from the notation frame.

# An Illustration: Sequential Data and Adaptive Strategies III

**Example 1.5 (book):** data from $\mathrm{N}(\theta^\dagger, 1)$ with true $\theta^\dagger = 0.3$; $H_0 : \theta^\dagger = 0$. Six strategies for choosing $\theta_i$: (i) true value $0.3$ (log-optimal); (ii) misspecified $0.1$; (iii) random, $\theta_i \sim \mathrm{Unif}[0, 0.5]$ i.i.d.; (iv) MAP with prior $\mathrm{N}(0.1, 0.2^2)$; (v) MLE from $X_1, \ldots, X_{i-1}$ (started at $\theta_1 = 0.1$); (vi) discrete mixture over a uniform grid on $[0, 0.6]$. For large $n$, methods (iv), (v), (vi) grow *almost as fast* as the (unrealistic, oracle) method (i); misspecified/random strategies (ii), (iii) grow more slowly.

# An Illustration: Sequential Data and Adaptive Strategies IV



# An Illustration: Sequential Data and Adaptive Strategies V

Our own simplified illustration in the same spirit: a single e-process run under $H_0$ (bounded, rarely crosses the $1/\alpha$ threshold) versus under the true alternative (grows essentially linearly in log-scale, i.e. exponentially).

# Python: 1000 Replications — Confirming Mean E-value $\leq 1$

```python
import numpy as np
rng = np.random.default_rng(3)

n, p0, q0 = 20, 0.5, 0.7
n_reps = 1000
S = rng.binomial(n, p0, size=n_reps)          # simulate data under H0, 1000 times
E = (q0/p0)**S * ((1-q0)/(1-p0))**(n-S)        # e-value for each replication

print("Sample mean of E over 1000 replications:", E.mean())
print("Sample max of E:", E.max())
print("Fraction with E > 1/0.05 = 20:", np.mean(E > 20))

# The one specific dataset we highlighted in the worked example:
S_obs = 14
E_obs = (q0/p0)**S_obs * ((1-q0)/(1-p0))**(n-S_obs)
print("E for our observed 14-heads-out-of-20 dataset:", round(E_obs, 4))
```

**Output (excerpt):** `Sample mean of E` $\approx$ `0.99` (consistent with the $\leq 1$ guarantee up to Monte Carlo error), `Fraction with E > 20` $\approx$ `0.005--0.01` ($\leq 0.05$, as Markov's inequality demands), and `E for our observed dataset = 5.1844`, matching the hand computation on the previous slide.

# 1.7 A Second Example: The Soft-Rank E-value

Many classical procedures (permutation tests, Monte Carlo tests) produce statistics that are *exchangeable* under the null (reminder: their joint distribution is unchanged by relabeling) and use the *rank* of the real statistic among simulated copies as a p-value. This section converts that idea into an e-value.

## Setup

*Intuition:* we have one real test statistic $L_0$ and $B$ "fake" copies $L_1, \ldots, L_B$ that look statistically identical to $L_0$ under the null (e.g., generated by random permutations). We want to turn the comparison "is $L_0$ unusually large?" into a valid e-value.

- $L_0$: statistic from the real data (larger $=$ more evidence against $H_0$).
- $L_1, \ldots, L_B$: $B$ additional statistics, exchangeable together with $L_0$ under $\mathcal{P}$.
- **Example source of exchangeability**: testing independence of paired data $(X_i, Y_i)$ via sample correlation $L_0$; permute the $X$'s randomly $B$ times to get $L_1, \ldots, L_B$.

# The Soft-Rank E-value Formula

## Definition (soft-rank e-value)

First transform $L_0, \dots, L_B$ into nonnegative scores $R_0, \dots, R_B$ that preserve their order and exchangeability (details next slide), then define

$$E = \frac{(B+1)R_0}{\sum_{b=0}^{B} R_b} \qquad \text{(with } E := 1 \text{ if } \textstyle\sum_b R_b = 0\text{).}$$

- $R_0$: the transformed score of the *real* data's statistic.
- $\sum_{b=0}^{B} R_b$: total score across the real statistic and all $B$ exchangeable copies.
- *Why this is an e-variable:* by exchangeability of $R_0, \dots, R_B$ under $\mathcal{P}$, one can show $\mathbb{E}^{\mathbb{P}}[R_0 \mid \sum_b R_b] = \frac{\sum_b R_b}{B+1}$, which after rearranging gives $\mathbb{E}^{\mathbb{P}}[E] = 1$ (exact).

# Example 1.7: A Concrete Score Transformation

## Example 1.7: one concrete transformation

For a constant $r > 0$ and $L_* = \min_b L_b$:

$$R_b = \frac{\exp(rL_b) - \exp(rL_*)}{r} \geq 0.$$

- As $r \to 0$: $R_b \to L_b - L_* \geq 0$ (a simple shift).
- If the $L_b$ are already nonnegative, the simplest valid choice is $r = 0$ with $L_* = 0$, i.e., **just use** $R_b = L_b$ **directly**.

# Concrete Numbers: Soft-Rank E-value on Our Coin Data

We instantiate the abstract machinery with our own 20-flip dataset (14 heads), using the **longest run of heads** as the test statistic (a classic way to check whether a sequence looks "too streaky" to be a fair, independent coin).

## Step-by-step calculation

- **Data:** `1101110101010101 0111` (14 heads, 6 tails, engineered with a run of 6 heads at the start).
- **Step 1:** compute $L_0$ = longest head-run in the original data $= 6$.
- **Step 2:** draw $B = 4$ independent random permutations of the same 20 flips; compute longest head-runs $L_1, \ldots, L_4 = 7,\ 5,\ 4,\ 5$.
- **Step 3:** since all $L_b \geq 0$, take $R_b = L_b$ (the $r = 0$ choice). So $R_0, \ldots, R_4 = 6, 7, 5, 4, 5$, summing to 27.
- **Step 4:** soft-rank e-value $E = \dfrac{(4+1) \times 6}{27} = \dfrac{30}{27} \approx 1.111$.
- **Step 5:** ordinary permutation p-value $P = \dfrac{\#\{b : L_b \geq L_0\}}{B+1} = \dfrac{2}{5} = 0.4$ (namely $b = 0, 1$ tie/exceed $L_0 = 6$).
- **Check the book's inequality** $P \leq 1/E$: indeed $0.4 \leq 1/1.111 = 0.9$. ✓

# Python: Computing the Soft-Rank E-value

```python
import numpy as np
def longest_run(seq, val=1):
    best = cur = 0
    for x in seq:
        cur = cur + 1 if x == val else 0
        best = max(best, cur)
    return best

seq = [1,1,1,1,1,1,0,1,0,1,0,1,0,1,0,1,0,1,1,1]  # 20-flip data, 14 heads
L0 = longest_run(seq)
rng = np.random.default_rng(11)
B = 4
Ls = [L0] + [longest_run(list(rng.permutation(seq))) for _ in range(B)]
R = np.array(Ls, dtype=float)          # r=0 choice: R_b = L_b (nonnegative)
E = (B + 1) * R[0] / R.sum()
P = np.mean(np.array(Ls) >= L0)        # ordinary permutation p-value
print("L_0..L_4 =", Ls, " E =", round(E,4), " P =", P, " P<=1/E:", P <= 1/E)
```

**Output:** `L_0..L_4 = [6, 7, 5, 4, 5]`, `E = 1.1111`, `P = 0.4`, `P<=1/E: True` $(0.4 \leq 0.9)$ – matching the hand calculation exactly.

# Why "Soft-Rank"?

- The usual permutation p-value only looks at the *rank* of $L_0$ among $L_0, \ldots, L_B$: $P = \sum_b \mathbf{1}\{L_b \geq L_0\}/(B+1)$.
- After the exponential-style transform, $1/E$ can be seen as a "smoothed" version of that rank – hence **soft-rank p-value** for $1/E$, and **soft-rank e-value** for $E$ (analogous to how "soft-max" smooths the maximum via exponentiation).
- The book shows algebraically that $P \leq 1/E$ always – the direct p-value is never larger than the soft-rank p-value, so at first glance there's no advantage to using $E$ for a *single* test.
- **But:** Chapters 8–11 show e-values enjoy real advantages over p-values specifically in multiple testing and building confidence intervals – advantages invisible when testing just one hypothesis.

# 1.8 Historical Context: Who Invented E-values? I

**Short answer: nobody, and everybody.** If $\mathcal{P} = \{\mathbb{P}\}$ is simple, every optimal e-value is (or is dominated by) a likelihood ratio – and likelihood ratios are $\sim$100 years old, central to both frequentist and Bayesian statistics. What's new is recognizing e-values as a *general, composite-hypothesis-friendly* concept and the coordinated choice of the term "e-value" itself (agreed jointly by researchers in 2020).

The book credits five central historical figures, in chronological order:

- **Ville (1939):** brought (nonnegative) martingales to the center of probability theory; showed any measure-zero event has a nonnegative martingale exploding to $\infty$ on it. Motivated by critiquing von Mises' theory of "collectives," not statistical inference per se.
- **Wald (1945):** invented sequential analysis; nonnegative martingales (esp. the likelihood-ratio process) are central to his sequential probability ratio test, though framed around a single pre-specified stopping time.

# 1.8 Historical Context: Who Invented E-values? II

- **Kelly (1956):** proposed *log-optimality* – betting to maximize the expected log-growth rate of wealth in repeated favorable games, linking gambling to Shannon's information theory. (Breiman, 1961, generalized this beyond binary outcomes.)
- **Robbins (1970, with Darling 1967):** defined confidence sequences and "power-one" tests; recognized that products of e-values yield nonnegative supermartingales, and pushed for *anytime-valid* inference rather than Wald's fixed stopping rule.
- **Cover (1984):** developed log-optimal portfolio theory for financial trading with regret bounds against the best fixed portfolio in hindsight – a close cousin of log-optimal e-values.

**Vovk** later spearheaded the modern *game-theoretic probability* program (from 1993 onward, with Shafer and others), culminating in the term "e-value" being jointly adopted by Vovk, Shafer, Grünwald, Ramdas, and Wang around 2020.

# Recent Progress: The 2018–2020 Explosion I

A remarkable concentration of independent, aligned papers appeared on arXiv within a two-year window:

- **2018 (Howard, Ramdas, McAuliffe, Sekhon):** e-processes via families of supermartingales + Ville's inequality $\Rightarrow$ nonparametric confidence sequences.
- **2018 (Kaufmann & Koolen):** mixture method for likelihood ratios in exponential families; sequential KL-divergence bounds.
- **2019 (Shafer):** expository case for betting-based scientific communication and log-optimality.
- **2019 (Shafer & Vovk, book):** game-theoretic foundations of probability and finance.
- **2019 (Grünwald, De Heide, Koolen):** simpler e-process definition (valid at any stopping time); reverse information projection; meta-analysis applications.

# Recent Progress: The 2018–2020 Explosion II

- **2019 (Vovk & Wang):** averaging is essentially the *only* admissible way to combine dependent e-values; e-to-p and p-to-e calibrators. This paper triggered the community-wide agreement on the term "e-value."
- **2019 (Wasserman, Ramdas, Balakrishnan):** "universal inference" – e-values from split-likelihood-ratio statistics, valid with zero regularity conditions.
- **2020 (Vovk, Wang, Wang):** admissible p-value merging must go through e-values (via optimal transport duality).
- **2020 (Wang & Ramdas):** e-BH, an e-value analogue of Benjamini–Hochberg, controlling FDR under *arbitrary* dependence.
- **2020 (Ramdas, Ruf, Larsson, Koolen):** necessary and sufficient conditions for admissible SAVI (sequential anytime-valid inference) tools.
- **2020 (Waudby-Smith & Ramdas):** testing-by-betting for bounded-mean estimation; universal representation of nonnegative martingales.

# Unfortunate Name Clashes: Other “E-values” in the Wild

The letter “e” was popular before this book – watch out for these unrelated uses of “E-value” in other literatures:

- **BLAST E-value** (bioinformatics): expected number of chance hits in a sequence database; roughly $E = -\log(1 - P)$ applied to a p-value – a different object entirely.
- **Sensitivity E-value** (VanderWeele & Ding, 2017, causal inference): the minimum confounding strength (on a risk-ratio scale) needed to fully explain away an observed association.
- **Bayesian E-value** (Pereira & Stern, 1999): the *epistemic* value $ev(H \mid X)$ in the Full Bayesian Significance Test – a Bayesian evidence measure, unrelated to this book's expectation-based definition.

Reassuringly, the terms “e-variable” and “e-process” have no known clashes – use them alongside “e-value” when precision matters.

# 1.9 Road Map of the Book I

The book is organized into three parts, building from foundations to advanced applications.

## Part I: Fundamental Concepts (Ch. 1–4)

- **Ch. 2 — Validity.** Properties of e-values under the null: Markov's inequality, calibrators, randomized tests; explicit e-variables in simple examples.
- **Ch. 3 — Efficiency.** Properties under the alternative: e-power, log-optimality, existence of powered e-values, mixture/plug-in methods.
- **Ch. 4 — Post-hoc decisions.** E-values' role when parts of the testing problem (level, decisions, or even data collection) depend on the data itself.

# 1.9 Road Map of the Book II

## Part II: Core Ideas (Ch. 5–9)

- **Ch. 5 — Universal inference:** a simple, general method for building e-variables for composite nulls/alternatives – the first known test for many "irregular" problems (though improvable).
- **Ch. 6 — Log-optimal e-variables:** a unique "numeraire" e-variable always exists for a fixed alternative, connected to the reverse information projection.
- **Ch. 7 — SAVI:** sequential anytime-valid inference – e-processes, testing by betting, optional stopping, Ville's inequality.
- **Ch. 8 — Merging e-values:** combining via averages (arbitrary dependence) or products (independence).
- **Ch. 9 — Compound e-values:** e-Benjamini–Hochberg for false discovery rate control.

# 1.9 Road Map of the Book III

## Part III: Advanced Topics (Ch. 10–16)

- **Ch. 10:** approximate/asymptotic e- and p-values.
- **Ch. 11:** combining dependent e-values is only admissible via (weighted) averaging.
- **Ch. 12:** combining dependent p-values (via e-values and optimal transport duality).
- **Ch. 13:** e-confidence intervals; e-Benjamini–Yekutieli for false coverage rate.
- **Ch. 14:** further e-value/p-value contrasts and dualities.
- **Ch. 15:** improved (sub-$1/\alpha$) thresholds under shape or dependence assumptions.
- **Ch. 16:** connections to risk measures; nonparametric tests for forecasts.

# 1.9 Road Map of the Book IV

## Four cross-cutting goals

1. **Understanding the foundations:** Chapters 2, 4, 10, 14.
2. **Constructing good/powerful e-values:** Chapters 3, 6, 15.
3. **Designing inference methods:** Chapters 5, 7, 16.
4. **Designing multiple-testing methods:** Chapters 8, 9, 11, 12, 13.

In the first two categories, the object of interest is the e-value *itself*; in the remaining two, the e-value is a means to a specific statistical end.

# Chapter 1 — Key Takeaways

- An **e-value** is a nonnegative statistic with expectation $\leq 1$ under the null – a valid bet against $H_0$; large values are evidence against the null (opposite convention from p-values).
- E-values and p-values are **statistical cousins**: convertible via $P = 1/E$ and $E = \mathbf{1}\{P \leq \alpha\}/\alpha$, but e-values uniquely support post-hoc levels, optional continuation (via simple multiplication), and unknown sampling schemes.
- The preferred optimality criterion is **e-power** $\mathbb{E}^{\mathbb{Q}}[\log E]$, forced by the algebra of repeatedly multiplying e-values together.
- The **betting interpretation** (Forecaster vs. Skeptic) is the intuitive engine behind the whole framework: e-variables are exactly the wealth-generating bets Forecaster would accept.
- Two worked templates recur throughout the book: the **likelihood-ratio e-value** (our coin example: $n = 20$, $14$ heads, $E \approx 5.18$) and the **soft-rank e-value** (permutation-style tests, our longest-run example: $E \approx 1.11$).
- Five names anchor the history: **Ville, Wald, Kelly, Robbins, Cover** – with Vovk, Shafer, Grünwald, Ramdas, and Wang driving the modern synthesis.
- The rest of the book builds outward from these definitions: validity (Ch. 2), efficiency (Ch. 3), post-hoc decisions (Ch. 4), and onward through sequential and multiple-testing applications.